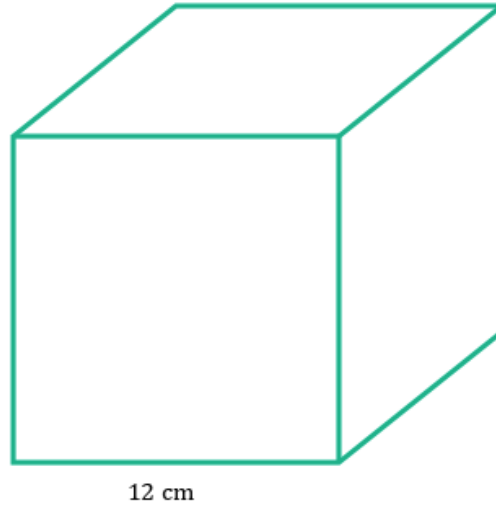


Volume of prisms and cylinders

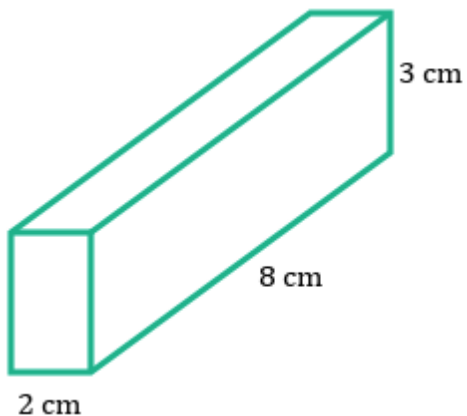


- 1 Find the volume of the cube shown.

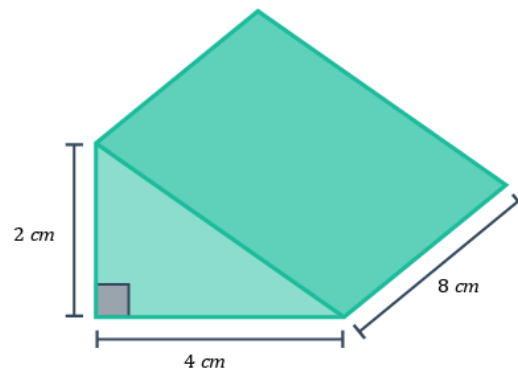


- 2 Find the side length of a cube with volume 27 cm^3 .
- 3 Find the length of the rectangular prism with volume 162 cm^3 , width 6 cm and height 3 cm .
- 4 A pentagonal prism has a volume of 176.4 m^3 and a cross sectional area of 29.4 m^2 . Find the height of the pentagonal prism.
- 5 Find the volume of the following prisms:

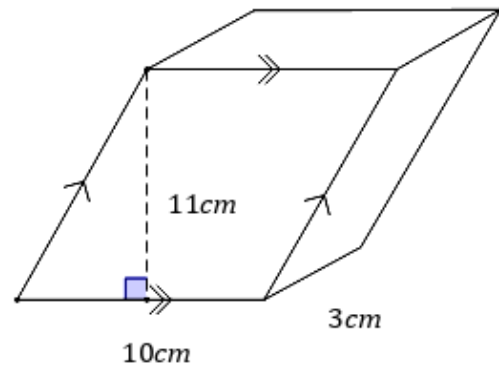
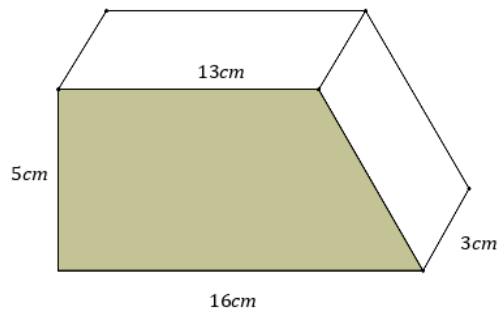
a



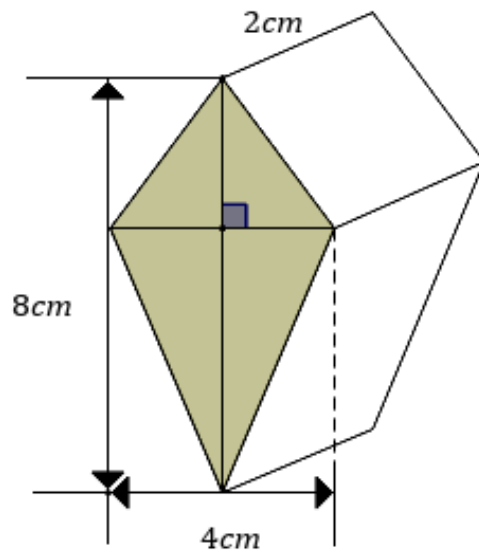
b



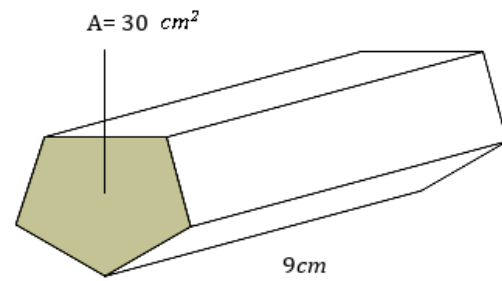
c



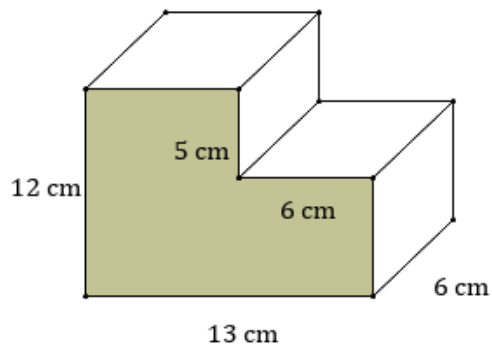
e



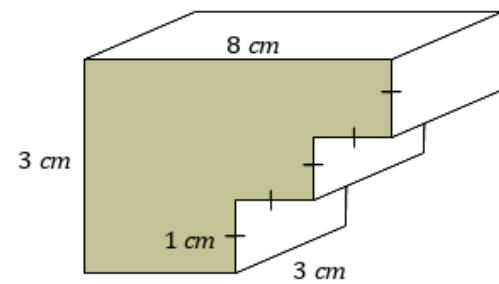
f



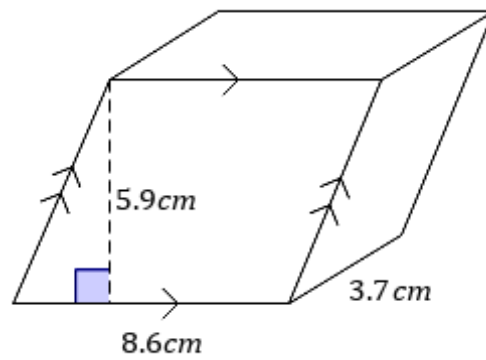
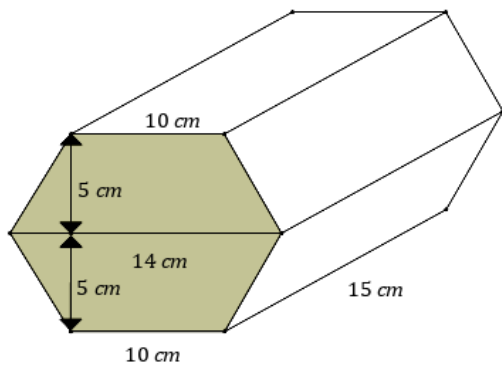
g



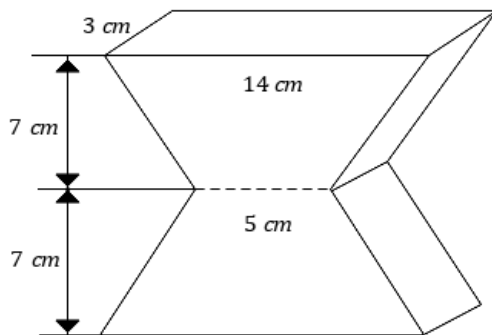
h



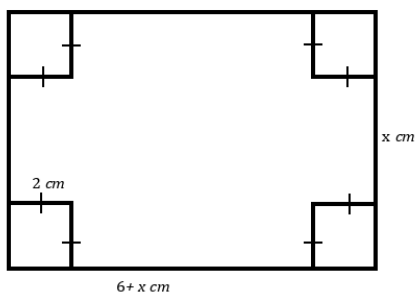
i



k



- 6 The length of a rectangular piece of cardboard is 6 cm longer than its width. Squares with sides of length 2 cm are cut from the four corners, and the flaps are folded upward to form an open box with a volume of 32 cm^3 . Determine whether following equations correctly describes the volume of the box.



a $x(x+6) = 32$

b $x(x+6)(2) = 32$

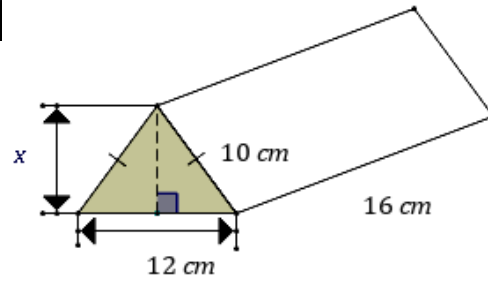
c $(x+2)(x-4) = 32$

d $(x+2)(x-4)(2) = 32$

7 Consider the following diagram.



- a Find the value of the unknown height x .
- b Hence find the volume of the solid.

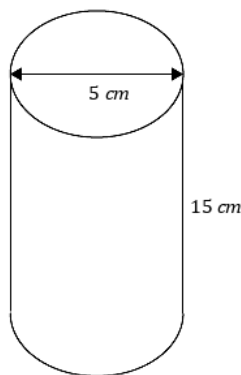


8 Find the volume of a cylinder correct to one decimal place if its diameter is 2 cm and its height is 19 cm.

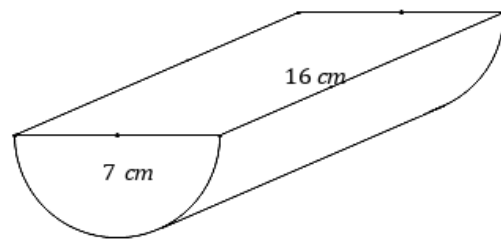
9 Find the volume of a cylinder with radius 7 cm and height 15 cm, correct to two decimal places.

10 Find the volume of the following solids, correct to one decimal place. All measurements are in centimeters.

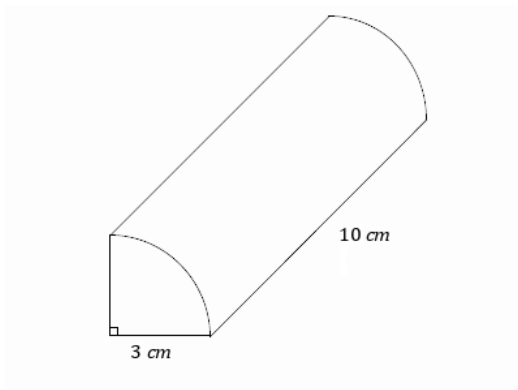
a



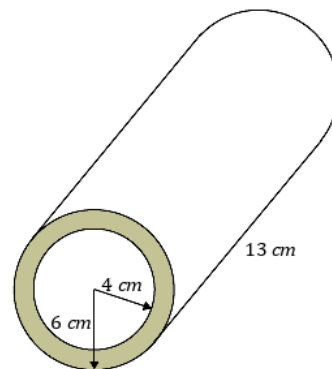
b



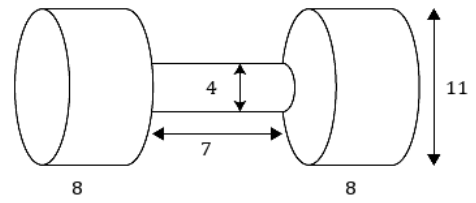
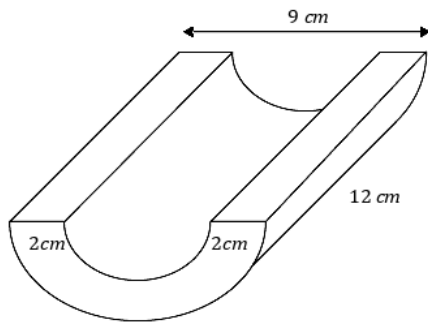
c



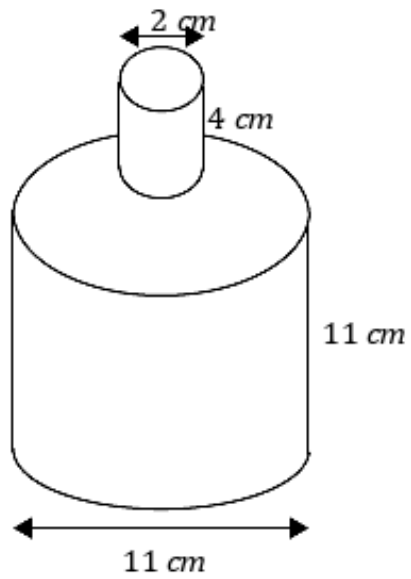
d



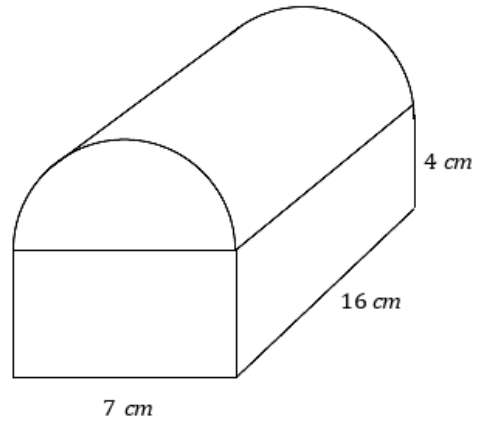
e



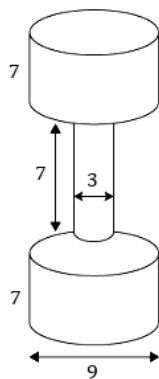
g



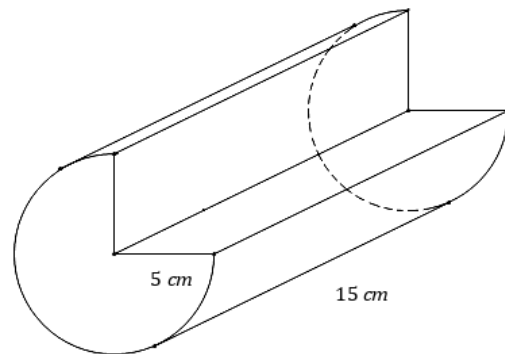
h



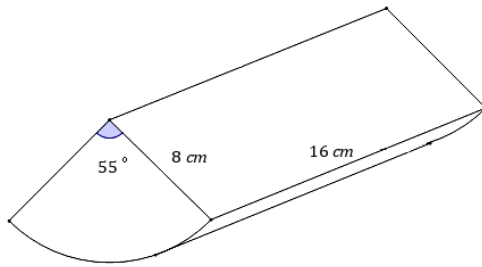
i



j

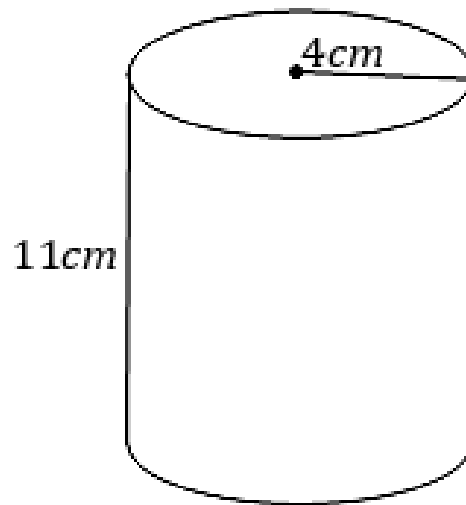


k

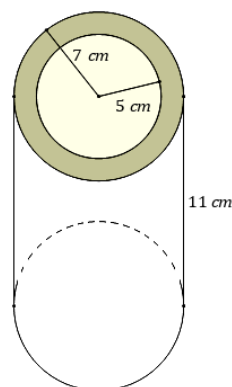


- 11 Xavier's mother told him to drink 4 glasses of water each day. She gave him a cylindrical glass with height 11 cm and radius 4 cm.

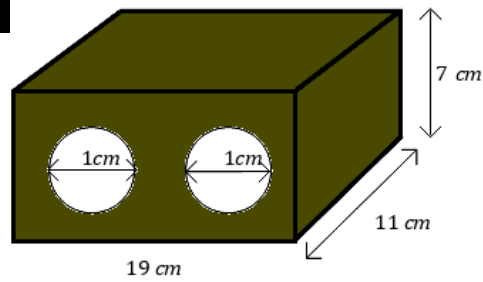
- Find the volume of the glass.
Round your answer to two decimal places.
- Assuming that he drinks 4 full glasses as his mother suggested, calculate the volume of water Xavier drinks each day.
Round your answer to two decimal places.
- If Xavier follows this drinking routine for a week, how many litres of water would he drink altogether?
Round your answer to one decimal place.



- 12 A cylinder is hollowed out by a hole of radius 5 cm. Find the volume of the remaining figure, correct to two decimal places.



- 13 We wish to calculate the volume of the solid shown by considering it as a prism. The cylinders are drilled perpendicular to the surface.



- a Calculate the area of the front face correct to two decimal places.
- b Hence find the volume of the prism.

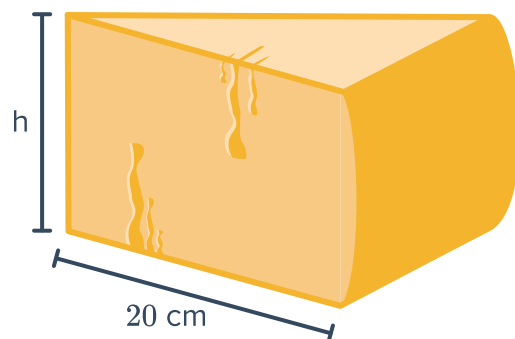
Additional questions

- 14 Queso Fresco is sold in wheels. The volume of a wheel of Queso Fresco is 2002 cm^3 : The height of a wheel of Queso Fresco is 6 cm, find the radius.



- 15 Parmigiano-Reggiano is formed in wheels, and then cut into twenty equal slices before being packaged and sold. If the volume of a certain wheel of Parmigiano-Reggiano is $8000\pi \text{ cm}^3$:

- a Find the volume of a slice of Parmigiano-Reggiano that has been packaged for sale.
- b Find the height of the wheel of Parmigiano-Reggiano.



- 16 A standard six-sided die has a side length of 5 mm. If Kyle keeps his die in a pouch with a volume of 1000 mm^3 , how many die can he fit in the pouch?

- 17 A chocolate company sells their bars in a special triangular prism packaging:



- Find the volume of the chocolate packaging.
- The chocolate only fills 85% of the packaging. Find the volume of one bar of chocolate.



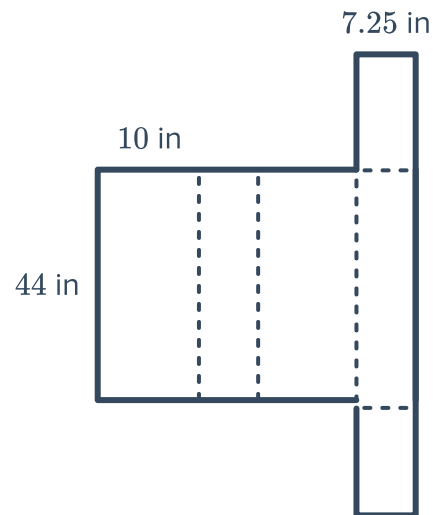
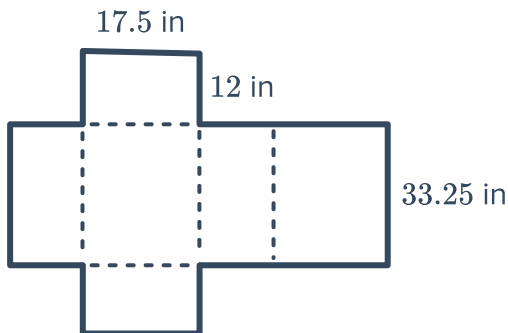
- 18 Tarpon Springs Aquarium is designing a new Reef Shark exhibit. In order to house the sharks they need a tank with a volume of 128 000 gallons.

In order to prevent the salt water from spilling over the edge of the tank, there needs to be a 1.5 meter clearance from the water level to the top of the shark tank. Using the conversion $1 \text{ m}^3 = 244 \text{ gal}$, design a rectangular tank that can hold 128 000 gallons of salt water.

- 19 U-Pack Moving Company sells two sizes of moving boxes. U-Pack claims to keep their prices lower than competitors by selling their boxes flat, and having customers assemble the boxes themselves.

Box 1 sells for \$2.29:

Box 2 sells for \$0.99:



If you have 60 ft^3 of stuff to pack into boxes for your move, which of the following combinations of boxes would you buy? Explain.

a $15 \times$ Box 1

a $36 \times$ Box 2